



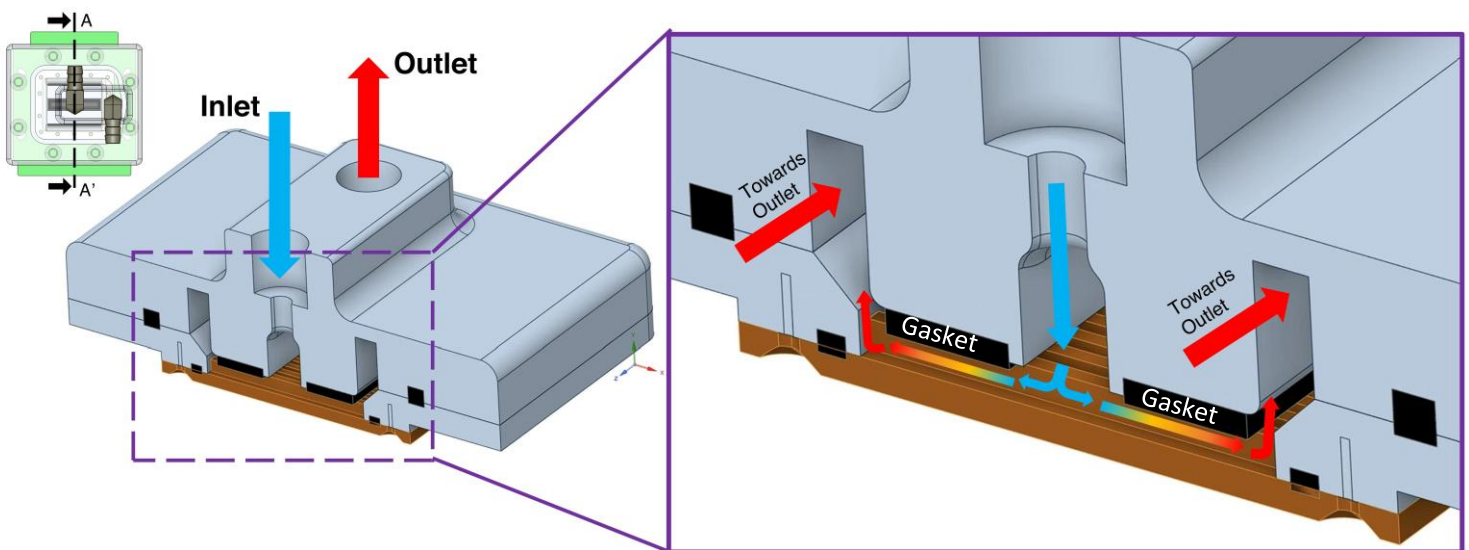
2025-2026 ASME K-16 / IEEE ITherm Student Cold Plate Design Competition

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For this competition, students will design, analyze, and optimize an **additively manufactured cold plate** to cool a computer chip via forced convection liquid cooling. Student teams will first design their cold plates and analyze them with a computational tool. Then, they will write a short white paper analyzing their designs and documenting the design process (due Jan 12, 2026). Student teams that have the highest predicted performance and most creative designs will have their designs by [Fabric8Labs](#) for experimental testing in the Spring. Based on the results, representatives from the best-performing teams will be invited to present their work at the [IEEE ITherm](#) conference in May, where a winner will be selected based on evaluation from a panel of experts.

Testing System & Manifold Overview

- Your simulated design must use the flow path geometry specified in the CAD files for the manifold and baseplate design without modifying dimensions. Only the fin plate region should be adjusted in your designs to meet the design goals.
- Download CAD drawings of the system from [Google Drive](#).



- $\frac{3}{8}$ " ID Tube Inlet and Outlet
- Split flow configuration
- 2mm thick Silicone gasket above fin plate to prevent flow bypass

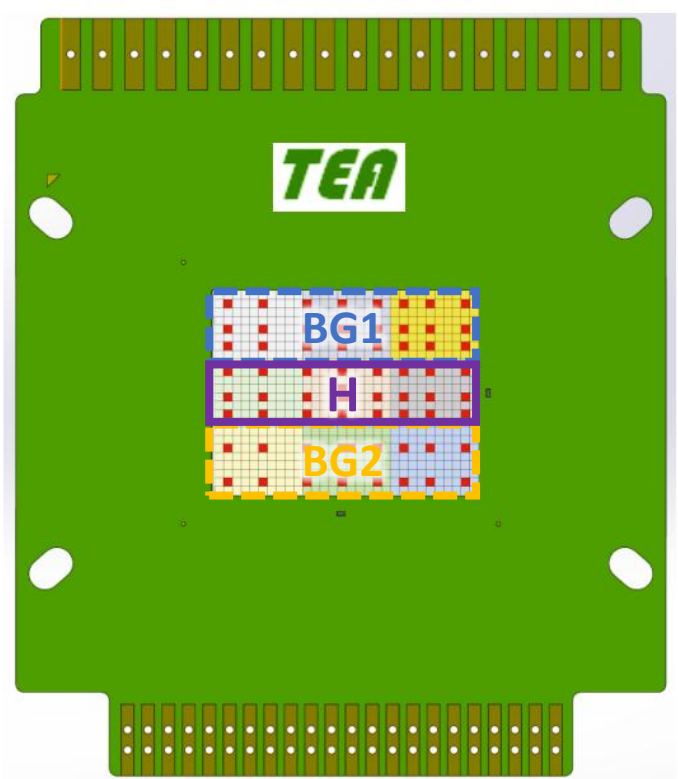
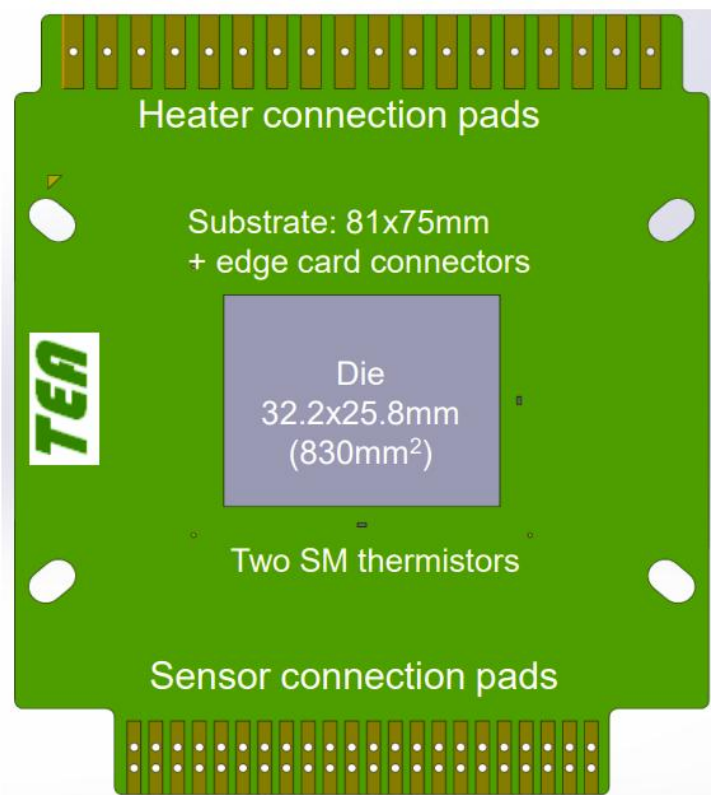
Simulation Inputs & Guidelines

Use the following parameters for simulations:

- **Coolant Fluid:** PG-25 (Polypropylene Glycol 25%)
- **Coolant Flow Rate:** 1.2 lpm
- **Inlet Temperature:** $T_{inlet} = 30^{\circ}\text{C}$
- **Ambient Temperature:** $T_{env} = 20^{\circ}\text{C}$
- **TTV-Cold Plate TIM Resistance:** $0.03 \text{ cm}^2\text{K/W}$
- **Power Map and TTV Configuration:** See Page 2 for details

Thermal Test Vehicle (TTV) Configuration

- The cold plate will be integrated with a Thermal Engineering Associates, Inc. (TEA) Thermal Test Vehicle TTV-4103.
- As shown in the image on the right below, there are 9 independent heating zones (color coded regions) and 68 temperature sensors (red squares).



- For your simulations, assume the following sections are heated:
 - **Zone H** [middle row of heaters (8.6 x 32.2 mm)]:
Total power of **400 W** uniformly distributed (1.44 W/mm²).
 - **Zone BG1** [top row (8.6 x 32.2 mm)]:
Total power of **200 W** uniformly distributed (0.72 W/mm²).
 - **Zone BG2** [bottom row (8.6 x 32.2 mm)]:
Total power of **200 W** uniformly distributed (0.72 W/mm²).
- The maximum temperature measured within each zone will be used when computing the Figure of Merit (see Page 3).



Design Objectives

Maximize the following **Figure of Merit (FoM)** based on the effective thermal resistance and pressure drop at the specified flow rate (1.2 lpm):

$$FoM = 0.7 \frac{R_{baseline} - R_{th,max}}{R_{baseline}} + 0.3 \frac{\Delta P_{baseline} - \Delta P}{\Delta P_{baseline}},$$

where the effective maximum thermal resistance is

$$R_{th,max} = \frac{\max(T_{chip}) - T_{inlet}}{q_{total}}$$

and the pressure drop is $\Delta P = P_{out} - P_{inlet}$ across the manifold inlet and outlet.

For the white paper, each team must determine $R_{baseline}$ and $\Delta P_{baseline}$ from a simulation using the provided baseline cold plate geometry following the same analysis as their design.

Scoring: White Paper

The objectives for the **white paper** are to demonstrate:

1. Effective use of thermal analysis and modeling to achieve a design with the highest Figure of Merit.
2. Effective and creative use of additive manufacturing.
3. Effective communication skills through a well-written, formatted, and clear white paper.

Metric	Assigned Point Value
Predicted FOM	25
Analytical Technique	30
Effective use of AM	30
Communication Skills	15

Designs failing to adhere to the additive design requirements as laid out in this document after the DfAM (Page 5) may not be selected for printing, despite high scores at the discretion of the organizing team and Fabric8Labs.

Final Competition Scoring

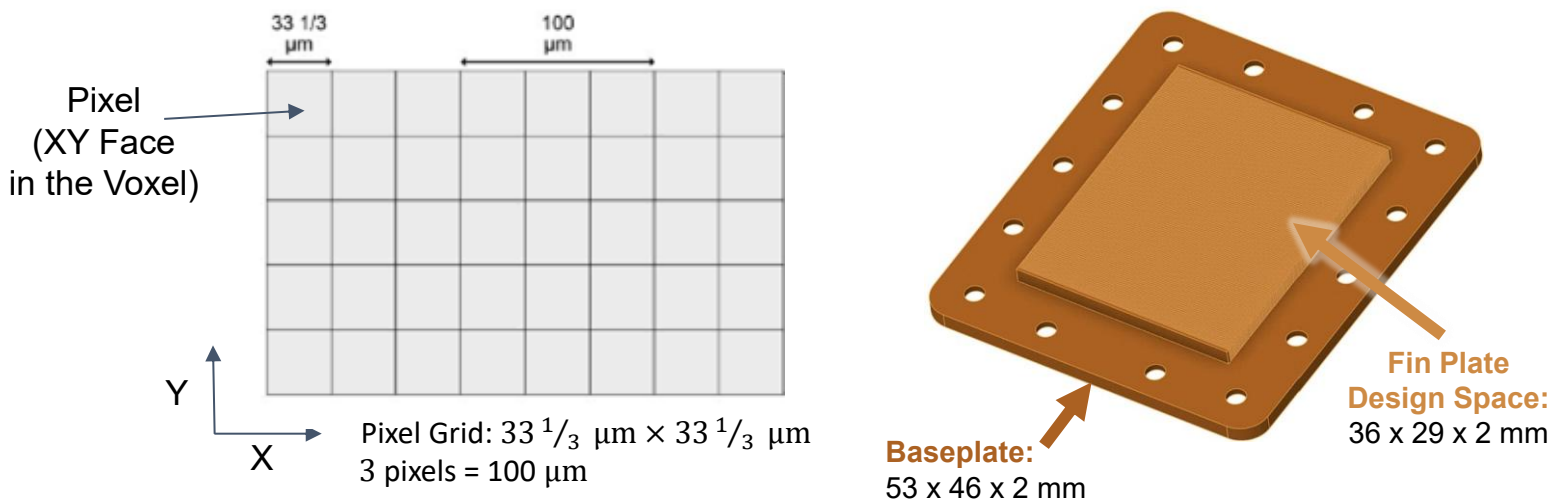
The objectives **for the final competition** are to demonstrate:

1. Effective use of thermal analysis and modeling to achieve a design with the highest Figure of Merit.
2. Effective and creative use of additive manufacturing.
3. Effective communication skills through an effective presentation of their design and results at *IEEE ITherm*.

Metric	Assigned Point Value
Experimental Figure of Merit	40
Effective use of AM	30
Presentation Skills	20

Design Space: Additive Manufactured Fin Plate

The design space is the **fin plate** shown in the diagram above. Your model must include the baseplate as shown in the diagram, but that must not be modified.



Fabric8Labs Additive Manufacturing Information

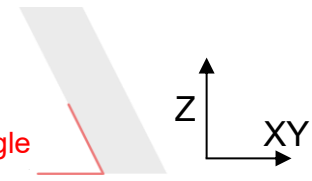
The cold plate will be 3D printed by [Fabric8Labs](https://fabric8labs.com) with pure copper. As printed, it is expected that the effective thermal conductivity is $\sim 380 \text{ W/(m K)}$. For modelling, it can be assumed that the surface roughness is 1.5 microns. Surface roughness can range depending on the geometry. The design and feature constraints to enable the printing are defined in the Design Guidelines section on the following page.

Minimum Feature Size & Design File Requirements

- You must submit a STEP file with your white paper that starts from the baseplate
- All STEP files are sliced with an XY voxel size of $33 \frac{1}{3} \times 33 \frac{1}{3} \mu\text{m}$ pixels. Therefore, X & Y dimensions of part features are defined by a grid of $33 \frac{1}{3} \times 33 \frac{1}{3} \mu\text{m}$ pixels.
 - Areas of the print that need high resolution, part feature size, and spacing should be defined in increments of $33 \frac{1}{3} \mu\text{m}$ in order to maintain feature fidelity
 - Example: A fin/gap of $133 \frac{1}{3} / 166 \frac{2}{3} \mu\text{m}$ will align to the pixels; whereas a fin/gap of $150 / 175 \mu\text{m}$ will not align to the pixels, causing actual fin feature thickness variation in the slices and the resulting print
- The Z dimension is defined by layer height. A minimum of 0.030 mm must be specified for competition.



Additive Manufacturing Design Guidelines

Feature	Size
Max Fin Plate Volume	$X = 36 \times Y = 29 \times Z = 2$ mm
Base Plate Size	$X = 53 \times Y = 46 \times Z = 2$ mm
Minimum Feature Size (positive or negative feature)	0.100 mm (3 pixels in X , Y , or Z direction)
Maximum Feature Size (positive feature)	X-Y : 0.500 mm (e.g., a straight fin down the length of the cold plate can be up to 0.5 mm in thickness)
	Z : 3 mm (limited by fin plate max dimension)
Minimum Feature Resolution (Voxel Size)	X-Y : $33 \frac{1}{3} \mu\text{m}$ (see previous page for detail)
	Z : $30 \mu\text{m}$ (see previous page for detail)
Minimum Feature Overhang Angle	20° (from XY plane)  Min feature overhang angle
Acceptable File Formats	.STEP [see page 4 for details]

Design Review for Manufacturability

Of the selected white paper designs, Fabric8Labs will perform design for additive manufacturing (DfAM) review. If required, Fabric8Labs will provide design change suggestions to improve the manufacturability of the parts. The DfAM review for manufacturability is envisioned to be collaborative in nature between the design team and Fabric8Labs.

Scoring Note: Effective Use of Additive Manufacturing

The scoring for effective of additive manufacturing score is an assessment by the reviewers of the white papers and panel of experts at the final competition of the overall creativity of the design and use of AM for achieve the design objective. Creativity is rewarded in this category along with the formation of novel structures that would otherwise not be possible with conventional manufacturing methods.



Competition Interest Form

Please fill out [this survey](#) ASAP to receive updates about the competition.

White Paper and Design/Simulation Submission Requirements

Each team must submit a [registration form](#) and upload the following files to [BOX](#) by the white paper deadline (Jan 12, 2026 at 11:59 pm US Eastern Time):

- **White paper (PDF)** following the provided template (see website)
- **CAD Files (STEP format required):**
 - **Base + Fin Plate Part File:** Your designed fin plate, including the baseplate
 - **Assembly:** Complete assembly file or set of files used for your simulations
- **Final Simulation Files (any appropriate program):** Upload a single zip file including a representative simulation of your final design configuration, including your calculated results. Make sure the file includes all the information to be opened on another computer. For example, in ANSYS, make sure to use the Export function rather than just sharing a saved file.

Team Guidelines

- **Eligibility:** The competition is open to both undergraduate and graduate students at universities around the world. Students on internship/co-op are eligible to participate, provided they have not yet graduated. Each team can submit one cold plate design, but more than one team is allowed for each university.
- **Team Size:** Teams can be of any size provided each team member's contributions to the creation of the design and writing of the white paper. Similar to authorship on a journal paper, each person should make a significant contribution to the project.
- **Mentors\Advisors:** While it is not required to have an official advisor, we encourage each team to identify a faculty member or industrial mentor who can provide feedback on their designs. However, we want to see the creativity, skills, and independent work of the students, not their mentors, in the designs.

Intellectual Property

The participants retain ownership, but the competition organizers and manufacturer (Fabric8Labs) shall be granted a perpetual, non-exclusive right to use the designs for educational, research, or commercial purposes.

Questions?

Please email the organizing team at asmeheatsinkchallenge@gmail.com with any questions about the competition.



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Ansys Software Access

If you need access to simulation software for this competition, ANSYS has graciously agreed to donate licenses to student teams without the node limitations of the free student version of their software.

To request licenses, one student per team should request access using this form: <https://ansysinc.formstack.com/forms/partner>. Please provide the following information in response to the questions about the competition.

Select your Partner Type: Student Teams

Student Team Type: Other

Please mention the name of competition(s) in which your team will be participating (include the year of participation): ASME K-16 / IEEE ITherm Student Cold Plate Design Competition

Organizing Team

Thanks to all the people who made this competition possible, including:

- Amy Marconnet (Purdue)
- Lang Yuan (Oracle)
- Han Hu (U. of Arkansas)
- Tianli Feng (U. of Utah)
- Vish Ganesan (UIUC)
- **Purdue Applied Research Lab (PARI):** John Holaday and Gabe Velarde
- **S-PACK Lab:** Tiwei Wei (UCLA), Akshat Patel, Yunchun Yang, and Ketankumar Yogi
- **Fabric8Labs:** Ian Winfield, Tanner Immonen, Gautam Gupta, Mike Matthew, and Jim Golden

Sponsors

Testing
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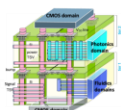


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Print Sponsor:



Simulation Sponsor:



Semiconductor Packaging & Thermal Lab

(Multiscale Materials, Heat Transfer, Mechanics, and Reliability in Semiconductors)

Sponsors make this competition possible.

**If you are interested in supporting the competition,
please email us at asmeheatsinkchallenge@gmail.com.**